IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

POWER SYSTEM PLANNING

Friday, 1 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : E. Spyrou
Second Marker(s) : M. O'Malley

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ANSWER ALL QUESTIONS

1. Variable renewable energy and storage

a) The average generation profiles for two potential wind farm investments, one onshore and one offshore, are estimated by measurements of wind speed over an one-year period and are given in Table 1.1. Their respective potential capacities are also given.

Table 1.1 Wind Energy Estimates

	Capacity (GW)	12AM to 4AM (MWh)	4AM to 8AM (MWh)	8AM to 12PM (MWh)	12PM to 4PM (MWh)	4PM to 8PM (MWh)	8PM to 12AM (MWh)
Onshore wind farm	2.5	3100	3050	3200	3300	3350	3200
Offshore wind farm	2	2800	2760	2900	3000	3050	2900

i) Estimate the capacity factor of the two potential wind farm investments. [2]

ii) Explain any differences in these capacity factors. [1]

iii) Estimate the yearly electrical energy that would be produced by each of these investments in TWh. [1]

iv) Stating any assumptions made, indicate which wind investment has the lowest Levelized Cost of Electricity (LCOE). [2]

b) The average generation profiles for two solar photovoltaic (PV) investments are given in Table 1.2. One is a solar farm investment and the other is a residential city-wide investment in solar PV for several hundred thousand rooftops. The profiles are estimated by measurements of solar irradiance over a five-year period.

Table 1.2 Solar PV Estimates

	Capacity (GW)	12AM to 4AM (MWh)	4AM to 8AM (MWh)	8AM to 12PM (MWh)	12PM to 4PM (MWh)	4PM to 8PM (MWh)	8PM to 12AM (MWh)
Solar PV Farm	10	0	2500	9400	11200	4000	500
Rooftop Solar PV	5	0	1600	3200	6000	1500	20

i) Estimate the capacity factor of the two potential PV investments. [2]

ii) Briefly explain any differences in these capacity factors. [1]

iii) Estimate the yearly electrical energy that would be produced by each of these investments in TWh. [1]

iv) Stating any assumptions made, indicate which solar PV investment has the lowest LCOE. [2]

c) Noting the fact that the measurements for wind and solar are for one and five years respectively (Table 1.1 & 1.2), briefly explain which estimates are likely to be more or less accurate. [2]

d) The potential onshore wind farm investment and the potential solar farm investment are in region A. Assume both these potential investments are made and there is no other generation in region A. The average demand profile for region A is given in Table 1.3.

Table 1.3 Region A demand profile

12AM to 4AM (GWh)	4AM to 8AM (GWh)	8AM to 12PM (GWh)	12PM to 4PM (GWh)	4PM to 8PM (GWh)	8PM to 12AM (GWh)
6	7.5	10.5	12	11.25	8.25

i) Estimate the electric energy demand in Region A in TWh. [2]

ii) Explain if there are any periods of renewable curtailment. [1]

iii) Explain if more generation investments are required to serve the demand reliably in region A. In one short sentence explain your answer. [2]

iv) There are three options for additional generation investment in region A: (1) Solar PV (2) Storage (3) Gas. If more generation investment is to be made in the region, assess each option and provide a very brief reason for why each option might be a good or bad investment. Note each investment option should be considered individually. [3]

e) The potential offshore wind farm investment and the potential rooftop solar PV investment are in region B. Assuming both these potential investments are made and there is no other generation in region B. The average demand profile for region B is given in Table 1.4.

Table 1.4 Region B demand profile

12AM to 4AM (GWh)	4AM to 8AM (GWh)	8AM to 12PM (GWh)	12PM to 4PM (GWh)	4PM to 8PM (GWh)	8PM to 12AM (GWh)
1.92	2.4	3.36	3.84	3.6	2.64

i) Estimate the electric energy demand in Region B in TWh. [2]

ii) Explain if there are any periods of renewable curtailment. [1]

iii) Explain in one short sentence if more generation investments are required to serve the demand reliably in region B. [2]

iv) There are three options for additional generation investment in region B: (1) Solar PV (2) Storage (3) Gas. If more generation investment is to be made in the region, assess each option and provide a very brief reason for why each option might be a good or bad investment. Note each investment option should be considered individually. [3]

v) Briefly explain if connecting region A and B is a good idea. [3]

2. Modelling demand for electric vehicles in a capacity expansion problem

a) In addition to traditional uses of electricity, reflected in $Demand_t$, in the basic capacity expansion model, electricity will be used in the future to charge the battery of electric cars. This new demand is described by timeseries EV_t.

i) Indicate which constraint you will modify in the basic capacity expansion model to represent this new demand. [2]

ii) Provide the formulation of the modified constraint. [2]

b) Assuming that charging can be delayed up to 4 hours compared to the time indicated in EV_t, modify the model to consider optimal charging.

i) Note any new decision variable(s). [2]

ii) Show any new and modified constraint(s). [2]

c) Assume that 10% of EV_t is willing to pay up to £1,000/MWh to charge.

i) Present modified and new constraints and define any new decision variables. [4]

ii) Provide a new objective function term that considers that part of the EV demand is not willing to pay more than £1,000/MWh to charge. [2]

d) A utility has solved two optimization problems to optimize generation costs. One problem considers that all 'plug-in' points are unidirectional (i.e., EV can charge from the grid) and the other considers them bidirectional (i.e., EV can charge from the grid and discharge to the grid). Table 2.1 reports the costs of these two problems. In both problems, there are 37 million 'plug-in' points.

Table 2.1 Annualized costs for a power system in £billion per year

	Problem A	Problem B
Generation costs	£41	£37
Cost of plug-in infrastructure	£4	£7

i) Indicate if the following statement is true or false: "Plan A has unidirectional 'plug-in' points". [2]

ii) Provide two reasons that support your answer to (i). [4]

iii) Estimate the cost of a bidirectional 'plug-in' point. [1]

iv) Estimate the cost of a unidirectional 'plug-in' point. [1]

e) In the plan with bi-directional 'plug-in' points of Table 2.1, only half of the points are used in both directions. However, the utility does not know which half is using the bidirectional capability. Assume a 50/50 chance for unidirectional or bidirectional use at each point.

i) Estimate the generation cost reduction per 'plug-in' point that uses the bidirectional capability. [2]

ii) Draw a decision tree that helps the utility decide whether to buy unidirectional or bidirectional equipment for each point. List the decision node(s), chance node(s), probabilities, and outcomes. [6]

iii) Estimate how much the utility is willing to pay for perfect information on each 'plug-in' point. [4]

3. Investment plans under uncertainty

a) A power system operator has simulated operations of four portfolios in 34 weather years. Table 3.1 reports unserved energy for these portfolios in 5 weather years. Assume that all 34 weather years are equiprobable and that the weather years not reported in Table 3.1 have zero unserved energy.

Table 3.1 Unserved energy (MWh) per year

Weather year	Portfolio 1	Portfolio 2	Portfolio 3	Portfolio 4
1984/85	2440	2800	0	0
1985/86	57040	57510	86810	75870
1996/97	0	13270	0	0
2000/01	450	420	280	0
2005/06	800	1650	0	0

i) Calculate the Expected Unserved Energy for each portfolio. [4]

ii) Calculate the Value at Risk for the Unserved Energy at 95% confidence for each portfolio. [4]

iii) Calculate the Conditional Value at Risk of Unserved Energy at 95% confidence for each portfolio. [4]

iv) Draw a plot with the Expected Value of Unserved Energy on x axis and the Value at Risk of Unserved Energy at 95% on y axis. Identify which portfolios are dominated alternatives, which are non-dominated alternatives, and draw the trade-off curve. [6]

v) Draw the cumulative risk profile of portfolios 3 and 4. [6]

vi) Explain if there is stochastic dominance between portfolios 3 and 4. [2]

b) Assume that the operator has calculated the unserved energy reported in Table 3.1 ignoring the transmission network.

i) Indicate which parameters and constraints from the "Handout from lecture notes" the operator needs to add to their model to consider the network. [2]

ii) Note which parameters or constraints the operator should modify when a line l is on outage. [2]

iii) Assume that during hour h, renewable energy is curtailed at node n. What is the shadow price of the supply-demand balance constraint at node n and hour h ? Justify your answer based on KKT conditions. [3]

Handout from lecture notes

1. **Basic capacity expansion formulation including investment and operational decisions.**

   Objective function

$$\min \sum_g (Fixed_Cost_g \cdot cap_g + \sum_t Variable_{Cost_{g,t}} \cdot gen_{g,t}) + \sum_t VOLL \cdot nse_t$$

   Decision variables:

   - nse_t: involuntary demand curtailment in MWh
   - cap_g: installed capacity in MW
   - $gen_{g,t}$: dispatch level of generator g at time t

   Constraints:

   Supply-demand balance: $\sum_g gen_{g,t} + nse_t = Demand_t \quad \forall t$

   Capacity constraint for each generator: $gen_{g,t} \leq AVAIL_{g,t} \cdot cap_g \; \forall g, t$

   Non negativity constraints: $nse_t,\ cap_g,\ gen_{g,t} \geq 0$

2. **Two power flow formulations**

   a) PTDF-based

   Objective function

$$\min \sum_g (Fixed_Cost_g \cdot cap_g + \sum_t Variable_{Cost_{g,t}} \cdot gen_{g,t}) + \sum_t VOLL \cdot nse_t$$

   Decision variables:

   - nse_t: involuntary demand curtailment in MWh
   - cap_g: installed capacity in MW
   - $gen_{g,t}$: dispatch level of generator g at time t

   Constraints:

   Supply-demand balance: $\sum_g gen_{g,t} + \sum_n nse_{n,t} = \sum_n Demand_{n,t} \quad \forall t$

   Transmission line limits:

$$P_l^{MIN} \leq \sum_n PTDF_{l,n} \cdot \left(-Demand_{n,t} + \sum_g df_{g,n} \cdot gen_{g,t} + nse_{n,t}\right) \leq P_l^{MAX} \; \forall l, t$$

   Capacity constraint for each generator: $gen_{g,t} \leq AVAIL_{g,t} \cdot cap_g \; \forall g, t$

   Non negativity constraints: $nse_t,\ cap_g,\ gen_{g,t} \geq 0$

b) B-θ

Objective function

$$\min \sum_g (Fixed_Cost_g \cdot cap_g + \sum_t Variable_{Cost_{g,t}} \cdot gen_{g,t}) + \sum_t VOLL \cdot nse_t$$

Decision variables:

- nse_t: involuntary demand curtailment in MWh
- cap_g: installed capacity in MW
- $gen_{g,t}$: dispatch level of generator g at time t
- $\theta_{n,t}$

Constraints:

Supply-demand balance: $\sum_g gen_{g,t} + \sum_n nse_{n,t} = \sum_n Demand_{n,t} \quad \forall t$

Supply-demand balance at nodal level:

$$\sum_g df_{g,n} \cdot gen_{g,t} + nse_{n,t} - \sum_l A_{l,n} \cdot \sum_{n'} B_l \cdot A_{l,n'} \cdot \theta_{n',t} = Demand_{n,t} \quad \forall n,\ t$$

Transmission line limits:

$P_l^{MIN} \leq \sum_n B_l \cdot A_{l,n} \cdot \theta_{n,t} \leq P_l^{MAX} \ \forall l, t$

Capacity constraint for each generator: $gen_{g,t} \leq AVAIL_{g,t} \cdot cap_g \ \forall g, t$

Non negativity constraints: $nse_t,\ cap_g,\ gen_{g,t} \geq 0$

3. **Big-M reformulation of bilinear terms**

Constraint with bilinear term in transmission expansion problem:

$\sum_n B_l \cdot A_{l,n} \cdot \boldsymbol{\theta_{n,t}} \cdot \boldsymbol{u_l} = flow_{l,t} \quad \forall l_{candidate}$

Reformulation:

$-M \cdot (1 - u_l) \leq flow_{l,t} - \sum_n B_l \cdot A_{l,n} \cdot \theta_{n,t} \leq M \cdot (1 - u_l) \quad \forall l_{candidate}$

M: big enough constant

4. **KKT conditions**

Stationarity: $\frac{\partial L}{\partial x} = \frac{\partial f_0}{\partial x} + \sum_{i=1:m} \lambda_i \cdot \frac{\partial f_i}{\partial x} + \sum_{i=1:p} v_i \cdot \frac{\partial h_i}{\partial x} = 0, \frac{\partial L}{\partial v} = 0$

Primal feasibility: $h_i(x) = 0, f_i(x) \leq 0$

Complementarity: $\lambda_i \cdot f_i(x) = 0$

Dual feasibility: $\lambda_i \geq 0$

5. Lecture 3 problem formulation

Objective function

$$\min \sum_{h \in H} \left(\sum_{g \in G} VarOM_g \cdot P_{g,h} + \sum_{r \in RES} VarOM_r \cdot R_{r,h} + VOLL \cdot NSE_h \right)$$

Constraints

Supply-demand balance: $\sum_{g \in G} P_{g,h} + \sum_{r \in RES} R_{r,h} + NSE_h = Demand_h \; \forall \; h$

Capacity constraint for thermal generator:

$0 \leq P_{g,h} \leq CAP_g \;\; \forall \; g, h$

Approximate inelastic operation of nuclear generation:

$P_{NUCLEAR,h} = CAP_{NUCLEAR} \;\; \forall \;\; h$

Capacity constraint for renewable generators:

$0 \leq R_{r,h} \leq CAP_r \cdot CF_{r,h} \qquad \forall \; r \; \in RES, h \; \in H$

Non-negativity constraint: $NSE_h \geq 0 \; \forall \;\; h$

6. Lecture 4 problem formulation

Objective function

$$\min \sum_{h \in H} \left(\sum_{g \in G} VarOM_g \cdot P_{g,h} + \sum_{r \in RES} VarOM_r \cdot R_{r,h} + VOLL \cdot NSE_h \right)$$

Constraints

Supply-demand balance:

$\sum_{g \in G} P_{g,h} + \sum_{r \in RES} R_{r,h} + \sum_{\boldsymbol{s} \in \boldsymbol{S}} \left(\boldsymbol{DISCHARGE_{s,h}} - \boldsymbol{CHARGE_{s,h}} \right) + NSE_h = Demand_h \; \forall \; h$

Capacity constraint for thermal generator: $0 \leq P_{g,h} \leq CAP_g \;\; \forall \; g, h$

Approximate inelastic operation of nuclear generation: $P_{NUCLEAR,h} = CAP_{NUCLEAR} \; \forall \;\; h$

Capacity constraint for renewable generators: $0 \leq R_{r,h} \leq CAP_r \cdot CF_{r,h} \qquad \forall \; r, h$

Limit of charge rate: $\boldsymbol{0 \leq CHARGE_{s,h} \leq CAP_s \qquad \forall \; s \; \in S, h \; \in H}$

Limit of discharge rate: $\boldsymbol{0 \leq DISCHARGE_{s,h} \leq CAP_s \;\; \forall \; s \; \in S, h \; \in H}$

Limit of non-synchronous generation (inspired by EirGrid)

$$\sum_{\boldsymbol{r \in RES}} \boldsymbol{R_{r,h}} + \left(\boldsymbol{DISCHARGE_{BESS,h}} - \boldsymbol{CHARGE_{BESS,h}} \right) \leq \boldsymbol{75\% \cdot Demand_h \; \forall \; h}$$

Evolution of state of energy:

$$\boldsymbol{SOC_{s,h} = SOC_{s,h-1} + CHARGE_{s,h} \cdot Eff_s - \frac{DISCHARGE_{s,h}}{Eff_s} \quad \forall \; s \; \in S, h \; \in H \backslash 1}$$

State of energy limits: $\boldsymbol{0 \leq SOC_{s,h} \leq SOC_s^{max} \; \forall \; s \; \in S, h \; \in H}$

End-effects for state of energy: $\boldsymbol{SOC_{s,1} = SOC_{s,t_{end}} = 50\% \cdot SOC_s^{max} \; \forall \; s \; \in S}$

Non-negativity constraint: $NSE_h \geq 0 \; \forall \;\; h$